

AI CONSULTING PROPOSAL · I&E LLC (FICTIONAL CLIENT)

KlarSchiff

From a prompt to a working pre-shipment
agent

Less searching, less stress, no surprises at the border. The AI does the checking.
People keep the decisions.



THE STORY IN SIX QUESTIONS

Goal, problem, approach, output, value and relevance

1 Problem and goal

Customs holds cost money and stress;
we catch errors at the desk

Slide 3

2 Approach

An AI strategy, a five-step agent,
official sources

Slides 4–6

3 Final output

A working app, its scope and its tested
quality

Slides 7–10

4 Business value

Break-even in month 7 (base case),
risks under control

Slides 11–12

5 Next steps and relevance

A 3-month pilot, then agile delivery over
about 12 months

Slides 13–14

6 Recommendation

Start small, prove it, then scale

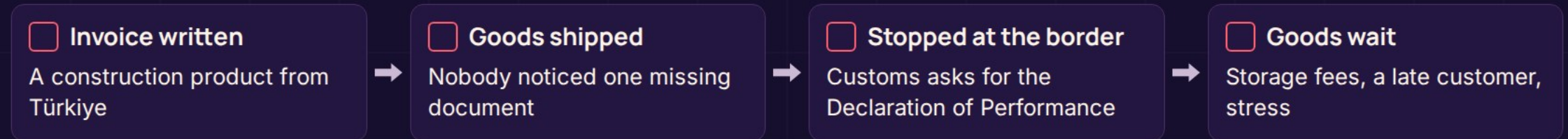
Slide 15

One idea behind every slide: the AI does the checking, people keep the decisions.

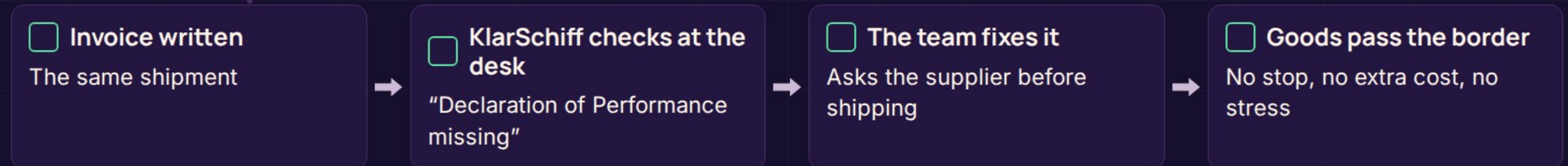
2 · PROBLEM · ONE EXAMPLE

The problem in one shipment, without jargon

TODAY



WITH KLARSCHIFF



HS code

The product's ID number for customs: 6 digits, used worldwide

Customs broker

The expert who files the customs declaration

Declaration of Performance

A certificate that EU construction products need

The AI strategy for I&E on one page

☐ 1 · Vision

Every shipment is checked before it leaves. The AI does the searching; people keep the decisions and get their time back.

☐ 2 · Use case choice

Chosen: pre-shipment check (high value, low risk). Ruled out: automatic customs filing (liability, trust) and delay forecasting (no historical data yet).

☐ 3 · Data and tools

Official sources (HS 2022, EU and US rules), the client's master list, GPT-4o-mini, EU-region tracing.

☐ 4 · People and change

A product owner at the client, 2 trained users, the broker as expert reviewer, AI literacy training.

☐ 5 · Governance and risk

EU AI Act: minimal risk, with high-risk-style controls. GDPR, decision log, confidence threshold 0.75.

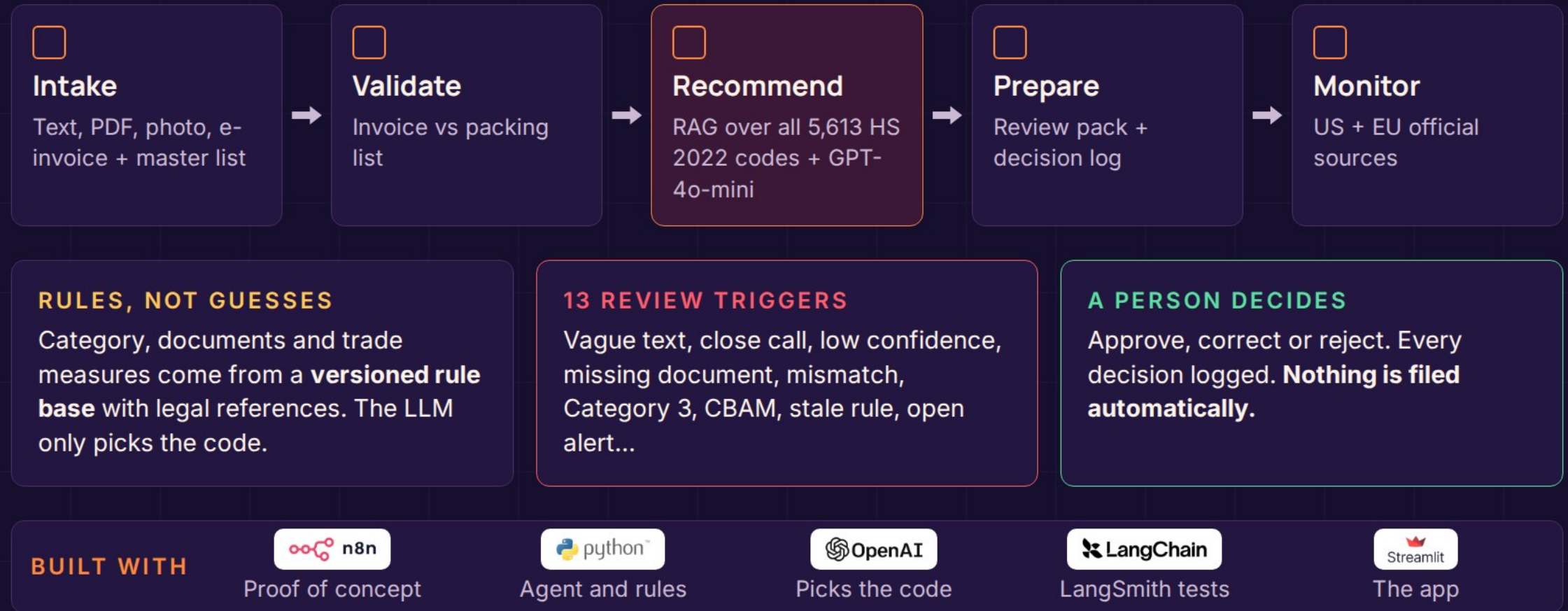
☐ 6 · Roadmap and KPIs

3-month pilot in 6 sprints, then go / no-go. –50% holds, review under 10 min, ≥90% codes accepted.

Start small, prove it, then scale: one route and one product family first; a wider rollout only after a measured "go".

3 · APPROACH · INSIDE THE AGENT

Five steps: the AI picks the code, rules decide the rest



The AI does not invent rules: every check has an official source

What is checked	Official source	Version / date
Customs code (6 digits)	UN Harmonized System 2022: 5,613 codes, public domain	HS 2022 · a code outside it is flagged
EU line (8 digits)	EU Combined Nomenclature 2026, official EU file	CN 2026
US line and duty	US International Trade Commission tariff (USITC API)	checked live for the shipment
Documents and trade rules	16 EU and US regulations, each with legal reference and source link	start date + last-verified date per rule
Changes	Daily monitor: EU pages + US Federal Register	alert → a person reviews the rule

Confidence below 0.75

→ a person reviews

Code not in the reviewed list

→ a person reviews

Duty rates

→ always the live official tariff

In one sentence: Every rule comes from an official source with a date. When something is unclear, a person reviews.

4 · FINAL OUTPUT · LIVE DEMO

Live demo: precast wall panels with a missing DoP



AI pre-shipment co-pilot · HS code suggestion, document check and tariff monitor · **Clear answers, human decisions.**

Suggest the customs code · Check the documents · Applies official rules · You decide

2 · Suggestion

A person must review this shipment

2 reason(s):

- Required document(s) missing: Declaration of Performance (DoP / DoPC)
- Offline mode (no LLM): keyword match only.

Money tip

Preference possible under the EU-Türkiye customs union: with the proof (A.TR movement certificate) the EU duty can be lower or zero. No proof found: ask the supplier before shipping, or the full duty applies. A person checks the origin rules; rate in TARIK for the shipment date.

Review ticket R-0001 opened · due 2026-10-05 · see the Review queue tab



1 · Intake
read



2 · Validate
0 issue(s)



3 · Recommend
HS 6810.91



4 · Rules
1 missing



5 · You decide
waiting for you

HS CODE

6810.91

Prefabricated structural components for building or civil...

CONFIDENCE

77%

Band: medium · below 75%: a person reviews

CATEGORY

2 / 3

CE-marked product

 Why  Documents  Rules  Sources & data

1 · INPUT

Invoice text, origin, destination, documents, optional CSV / PDF / e-invoice

2 · RESULT

HS 6810.91 · Category 2 · **DoP missing** → manual review

3 · DECISION

Approve / correct / reject → decision log → review pack for the broker

ALSO IN THE APP

E-mail inbox · Ask KlarSchiff · money tip (A.TR) · review queue · dashboard

What the MVP does today, in practice

INPUT

☐ Typed text

☐ PDF invoice

☐ Photo or scan

☐ CSV lines / batch

BAD DOCUMENT? WHAT HAPPENS

Poor scan, unreadable text	→ low confidence → a person reviews
Missing document or field	→ listed as missing, never green
Invoice ≠ packing list	→ both values shown → a person decides
Other language	→ translated; original kept; always reviewed
Handwriting	→ not tested yet → always reviewed

IN SCOPE TODAY

- 6-digit code with reasons and sources
- Documents and trade rules check
- English / German screen
- Master list, batch check, tariff monitor
- Decision log for audits

NOT IN THE MVP

- Filing the declaration (the broker does it)
- Connection to ERP / TMS (after a pilot "go")
- 10-digit EU code (TARIC): pilot work

4 · FINAL OUTPUT · FOR GERMAN TEAMS

Ready for German teams and invoices in other languages

A TURKISH INVOICE

KlarSchiff
AI pre-shipment co-pilot - HS code suggestion, document check and tariff monitor - Clear answers. Human decisions.

Suggest the customs code | Check the documents | Apply official rules | You decide

2 · Suggestion

A person must review this shipment
3 reason(s):

- Text in 'tr' and no AI model available to translate it.
- Category 3: additional trade measures or controls always need a person.
- Offline mode (no LLM): keyword match only.

Money tip
Preference possible under the EU-Türkiye customs union: with the proof (ATR movement certificate) the EU duty can be lower or zero. No proof found: ask the supplier before shipping, or the full duty applies. A person checks the origin rules; rate in TARIC for the shipment date.

Review ticket R-0001 opened - due 2026-10-05 - see the Review queue tab

1 · Intake read
2 · Validate 0 issue(s)
3 · Recommend HS 2523.29
4 · Rules 6 to confirm
5 · You decide waiting for you

HS CODE
2523.29
Other Portland cement

CONFIDENCE
75%
Band: medium - below 75%: a person reviews

CATEGORY
3 / 3
Trade measures or controls

Warum | Unterlagen | Regeln | Quellen & Daten

THE SAME APP IN GERMAN, ONE CLICK

KlarSchiff
KI-Co-Pilot vor dem Versand - HS-Code-Vorschlag, Dokumentenprüfung und Zollmonitor - Klare Antworten. Menschliche Entscheidungen.

Vollständig den Zolltarif vor | Prüfe die Unterlagen | Wende offizielle Regeln an | Sie entscheiden

2 · Vorschlag

Eine Person muss diese Sendung prüfen
2 Grund/Gründe:

- Pflichtunterlage(n) fehlen: Leistungserklärung (DoP / DoPC)
- Offline-Modus (kein Sprachmodell): nur Stichwortsuche.

Spartipp
Präferenz möglich (Zollunion EU-Türkei): mit dem Nachweis „Warenerkläreinheitsbescheinigung ATR“ kann der EU-Zoll niedriger oder null sein. Kein Nachweis gefunden: vor dem Versand beim Lieferanten anfordern, sonst gilt der volle Zollsatz. Eine Person prüft die Ursprungsregeln; Zollsatz in TARIC zum Versanddatum.

Prüfticket R-0001 eröffnet - fällig 2026-10-05 - siehe Reiter Prüfliste

1 · Eingang gelesen
2 · Abgleich 0 Abweichungen
3 · Vorschlag HS 6810.91
4 · Regeln 1 fehlt/fehlen
5 · Sie entscheiden wartet auf Sie

HS-CODE
6810.91
Prefabricated structural components for building or civil ...

SICHERHEIT
77%
Stufe: mittel - unter 75%: eine Person prüft

KATEGORIE
2 / 3
CE-gekennzeichnetes Produkt

Warum | Unterlagen | Regeln | Quellen & Daten

English / German

One click; the code and the rules stay the same

Turkish and Chinese

Detected, translated by the AI, original kept, always reviewed

E9

Found in testing: a Turkish photo was labelled English. Fixed and re-tested.

Tested in LangSmith: good scores, real errors, honest limits



LangSmith, by LangChain

20 test cases · LangSmith	v2.0	v2.1
HS code (6-digit)	0.95	0.95
Missing documents flagged	1.00	1.00
Routed to a person when needed	0.95	1.00
No false all-clear	0.95	1.00

Plus 17 cases from other industries: code right in 3 of 4, every unclear case sent to a person.

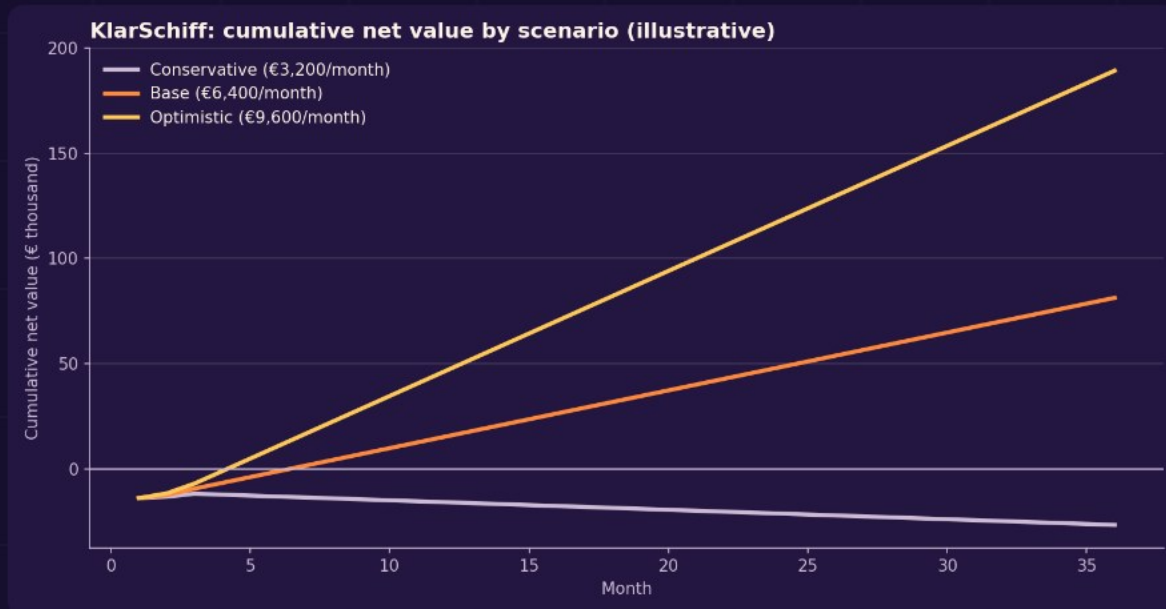
REAL ERRORS WE FOUND AND FIXED

- E7** A vague shipment passed with no review
Fix: vague text → a person
- E8** Right translation, wrong code
Fix: close call → a person chooses
- E9** A Turkish photo was labelled English
Fix: own language check, re-tested
- E10** Suggested a code that does not exist
Fix: codes outside HS 2022 are flagged
- E11** "Leistungserklärung fehlt" (DoP missing) was not seen
Fix: the check reads the whole word, re-tested

In one sentence: The scores are good but probably optimistic, so the pilot starts with a blind test.

5 · BUSINESS VALUE

Base case: break-even in month 7, +60% after three years



Scenario	Value	Break-even	ROI 12 / 36 m
Conservative	€3,200 / month	no break-even at €3.5k	-34% / -20%
Base	€6,400 / month	month 7	+32% / +60%
Optimistic	€9,600 / month	month 5	+99% / +141%

More honest than Round 1: "1.6 months" ignored the subscription. With all costs: month 7.

Lesson: the subscription follows measured value (€2k tier → conservative case breaks even in month 15).

Costs: pilot €12.5k + training €1.3k, subscription €3.5k/month from month 4, technology €150/month (LLM tokens under €5). Values illustrative.

Risks under control: minimal risk under the EU AI Act

TOP RISKS (SCORE = PROBABILITY × IMPACT)

- 16** **Tariff or rule change not noticed**
Control: monitor, alerts, dates
- 15** **Wrong HS code accepted**
Control: a person approves every code
- 12** **Automation bias**
Control: override rate, spot audits
- 12** **Personal data sent to an AI provider**
Control: only goods data, DPA, EU region
- 10** **Liability for customs errors**
Control: the declarant stays responsible

EU AI ACT

AI system?	Yes
Prohibited practice?	No
High-risk use?	No: no decisions about people
Result	Minimal risk

GDPR

Only goods descriptions go to the AI · data processing agreements · EU region · logs per team, not per person

6 · NEXT STEPS

A 3-month pilot in six 2-week sprints, then a go / no-go



Why now: CBAM since Jan 2026, a new EU steel measure since Jul 2026, deforestation rules from Dec 2026, e-invoices 2027–28. Manual checks get harder every year.

Agile delivery with Scrum: about 12 months to full production



Why 12 months: brokerage AI pilots run 10–12 weeks, enterprise AI rollouts 6–18 months, and only 54% of AI projects reach production (Gartner). Method: Scrum Guide 2020.

RECOMMENDATION

Start a 3-month paid pilot: one route, one product family, a blind test first.

Time back

Review from about 40 minutes to under 10 per shipment (pilot target).
Time for the work that needs people.

Lighter days

Fewer surprises at the border, fewer urgent calls, less stress for the team.

Confident decisions

Every answer comes with its source.
A person always decides.

A good agent does more than think: it saves time (no repeated work), saves money (it finds lower duties) and gets better with every correction.

KlarSchiff: clear answers, human decisions.